Exploring Urban Patterns in NYC Through Taxi Activity

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Abstract*—This study investigates whether High Volume For-Hire Vehicle (HVFHV) taxi data can serve as a proxy for various socioeconomic patterns across New York City, addressing three specific questions: whether taxi activity correlates spatially with crime, whether fare and tipping behaviour can be predicted from trip-level features and whether changes in taxi activity signal neighbourhood gentrification. Multiple analyses were conducted by merging millions of NYC Taxi & Limousine Commission trip records with NYPD arrest data, US Census indicators and NYC Department of Building permit issuance data. Fare prediction achieved a $MAE = 5.09$ and $R^2 = 0.82$ with XGBoost, with trip distance and trip duration as the dominant features. The research further investigated tip prediction, which proved harder due to the absence of cash tips in the dataset ($F1 = 0.29$). A modest positive correlation was identified between late-night taxi drop-offs and disorderly conduct arrests. Speaking more broadly, drop-off demand was generally more correlated with arrests than pickup demand. The analysis also found that taxi activity was more correlated with non-white-collar crime ($r = 0.417$) than white-collar crime ($r = 0.333$), reinforcing the view that taxi demand is more associated with street-level offences. Gentrification analysis found weak positive correlations between taxi revenue and changing house prices in more central zones, with building permit data corresponding to observed shifts in property investment (especially from Midtown Manhattan to Red Hook, Brooklyn). Even when no observed relationship between taxi patterns and gentrification was found, higher taxi revenues still occurred in some areas where there were greater levels of gentrification. Once combined, these findings suggest that HVFHV data can provide a valuable but partial lens into urban socioeconomic patterns, which highly depends on the examined area.

I. Introduction

New York City (NYC) has one of the world's most intense taxi environments, averaging more than 750,000 trips per day in 2025 [1]. Taxi trip records have high temporal resolution and detailed geographic information, which offers insight into human movement patterns within the urban environment. Taxi data in NYC has often been used as a proxy for human activity in the past few decades [2], reflecting patterns of work and leisure as well as underlying population densities.

In addition to the famous yellow taxi, first introduced in 1907, NYC has launched two other services: green 'boro' taxis in 2013 and High Volume For-Hire Vehicle (HVFHV) service in 2018, from Uber and Lyft, in an effort to cap the increasing market share they had acquired. The HVFHV dataset is used throughout this project as it offers the broadest and most consistent coverage of the modern for-hire vehicle market across all five boroughs.

Table I: Overview of datasets used in the project.

Dataset	Main Columns	Entries
NYC TLC High-Volume Trip Data [5]	Pickup time, pickup and drop-off zones, trip miles, trip time, provider, base passenger fare, tips, shared-trip flag	239,869,369
NYPD Arrest Data [6]	Date, arrest ID, offence description, coordinates	278,953
Median House Price [7]	ZIP zone, median house price	211
Mean Income [7]	ZIP zone, mean income	211
Educational Attainment [7]	ZIP zone, people with bachelor's degree	211
DOB Permit Issuance [8]	Zip code, issuance date, permit type, latitude, longitude	3,986,912

Previous work has explored the connection between mobility data and economic activity [3], as well as between mobility and social patterns [4]. Separately, much research has been conducted that examines crime distribution and socioeconomic inequality in cities. To our knowledge, however, limited prior work has jointly analysed all three of taxi activity, crime distribution and socioeconomic inequality in NYC. This study addresses this through the combination of HVFHV data with New York Police Department (NYPD) arrest records, US Census Bureau socioeconomic indicators and US Department of Building permit data to answer these three questions: (1) is taxi activity spatially correlated with crime? (2) can fare and tipping behaviour be predicted from trip-level features? and (3) can changes in taxi activity serve as a signal of neighbourhood gentrification?

All data used throughout this project are anonymised at source and do not identify any user or entry. The TLC trip data do not contain any passenger or driver identifiers, with the locations generalised using the zone-level pickup and drop-off IDs. The NYPD arrest data carry a higher risk of identification due to the inclusion of the unique coordinates of each arrest as well as age group, ethnicity, gender and a detailed charge description, though no individual can be identified from this dataset alone.

II. Data Preparation

A. Dataset Descriptions

1) NYC TLC High-Volume Trip Data: The primary dataset, Table I, was collected by the NYC Taxi and Limousine Commission (TLC) and covers all High Volume For-Hire Vehicle (HVFHV) trips across NYC. Each row represents one unique journey, such that cases where multiple passengers shared a vehicle to different destinations (e.g. via the UberX shared-ride service) are counted separately.

2) NYPD Arrest Data: The arrest dataset, taken as a proxy for crime, was sourced from NYC Open Data [6] and contains every arrest made across the five boroughs, regardless of whether the person was later charged.

3) Census and DOB Permit Issuance Data: The US Census Bureau was used as a source of three socioeconomic features, collated by ZIP Code Tabulation Areas (ZCTAs), for 2019 and 2025: the mean household income, median house value and the number of people with a Bachelor's degree. Median values were preferred where possible due to a small number of disproportionately large values assigned to properties and incomes. However, only mean household values by zone are available for the latter.

These measures were selected to form an instrument to gauge the potential gentrification of an area. As outlined in research concerning gentrification, income, property value and education level are all often used as indicators of gentrification taking place [9][10]. Alongside these, building permit issuance data, sourced from the NYC Department of Building (DOB), are used to track real estate investment in an area [8].

B. Data Cleaning

1) TLC Trip Data: The taxi dataset was cleaned to improve consistency and remove invalid records prior to analysis. The schema was standardised so that all records followed the same structure: text fields were cleaned by removing extraneous whitespace, null and empty values were converted to proper missing values and the licence IDs and flag fields were mapped to uppercase. Relevant fields, such as location IDs, trip duration, distance and all monetary values (fares, fees, tips and driver pay), were converted into appropriate numeric formats with any missing values set to zero before validation.

Logical checks were then applied, with timestamps verified to follow a valid order (request, pickup, on-scene, drop-off) and distance and monetary values checked for non-negativity. Rows with missing or incorrect required fields were removed and stored separately for reference. Outliers were excluded using fixed limits to avoid unrealistic data, and flag fields were standardised to `Y`/`N`. After cleaning, approximately 98.5% of records were retained.

2) NYPD Arrest Data: The arrest dataset was stripped of records outside the study's temporal window (2025). Rows with missing values in key analytical fields (specifically the arrest date and geographic coordinates) were removed. No records were excluded on the basis of demographic fields such as gender or ethnicity as these are not used in the analysis. Arrests were then grouped into broader categories, including labels for white-collar and blue-collar crimes, to aid in zonal comparison.

3) Census and DOB Permit Issuance Data: The census and building permits datasets required minimal cleaning as all columns relevant to this study's analysis were complete.

C. Spatial Harmonisation

1) Crime Data: The arrest dataset records each event as co-ordinates, with latitude and longitude, whereas the taxi dataset uses TLC zones. Thus, to enable joint analysis, each arrest was mapped to its corresponding TLC zone using QGIS with shapefiles of the TLC zones [11]. The final processed crime dataset hence included columns for the offence description, arrest date and TLC zone ID.

2) Census Data: The socioeconomic indicators from the census data were collated by ZCTA code zones, which do not align with the boundaries of TLC zones. As there are areas of overlap between ZCTA and TLC zones which cannot be remedied using a simple nearest-neighbour assignment, areal interpolation with area-weighted averaging was implemented to address this, redistributing values from source zones into target zones in proportion to the share of overlapping area [12].

The weighted average for a specific TLC zone (V_{TLC}) is defined as:

$$V_{TLC} = \frac{\sum_{i=1}^{n}(V_{ZIP,i} \cdot w_i)}{\sum_{i=1}^{n} w_i} \tag{1}$$

where $V_{ZIP,i}$ is the value from the i-th overlapping ZCTA and w_i is the area ratio (the area of intersection divided by the source area).

This approach is highly efficient for spatial analysis. By utilising area ratios and then normalising rather than simple centroid assignments, the model accounts for the actual geographic footprint of the data. This prevents "all-or-nothing" errors where a ZCTA might be assigned to a zone it only minimally overlaps, ensuring that gentrification metrics remain robust across the administrative areas.

Comparison of Figure 1 and Figure 2 demonstrates the success of this approach as the patterns present in Figure 1 are conserved as the values are mapped onto the new zones. This preserves the validity of any conclusions made, based on the new mapping.

Finally, the processed data had columns for the TLC zone and, for years 2019 and 2025, the median house value, mean income and the number of people with a Bachelor's degree.

D. Temporal Harmonisation

Given the scale of the HVFHV dataset, processing everything efficiently across multiple years was computationally infeasible. Thus, the analysis was restricted to a single year, 2025. This provided a manageable, yet still representative, sample of data that kept modelling and analysis development feasible. With 2025 chosen, the NYPD arrest data was

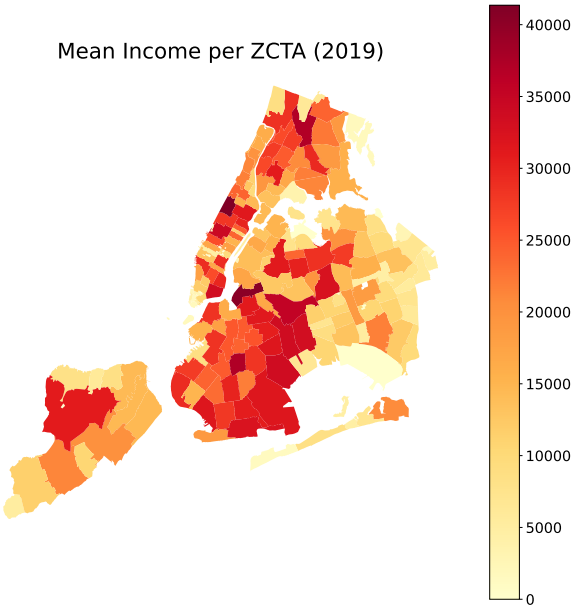


Figure 1: Heatmap of mean income per ZCTA (2019).

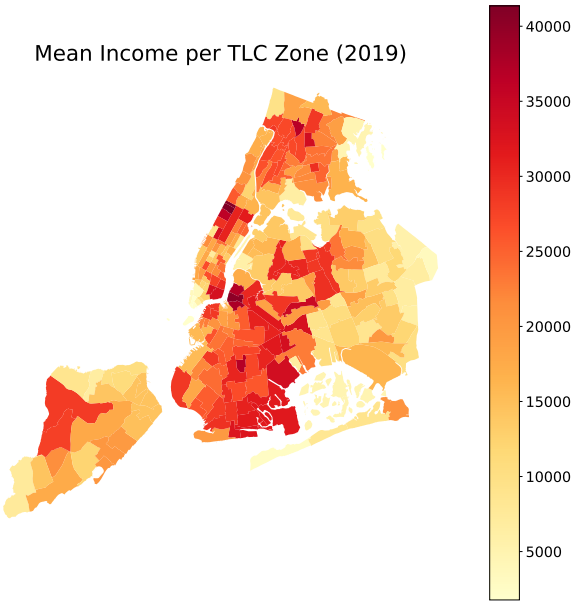


Figure 2: Heatmap of mean income per TLC zone (2019).

similarly filtered to allow analysis of taxi activity and crime frequency under the same urban conditions.

For the Crime-Taxi correlation analysis, both datasets were aggregated into a monthly time series at TLC-Zone level. This was necessary as using year-time series to analyse the data would leave too little temporal variation to study and produce a satisfying result. At the same time, a finer temporal resolution such as daily observation would have produced noisy series. Monthly aggregation thus provided a practical middle ground where it reduced noise and preserved meaningful temporal variation.

Comparison between these isolated data and those of 2019 was chosen to best observe gentrification as this requires sufficiently large a scale so as to assess its extent. This span of six years is the largest possible as HVFHV data were only collected in their current format from February 2019 onwards. Hence, since TLC data for January 2019 are missing, the sampling window was shifted to cover February 2019 to February 2020. So as to reflect this, sampled data for the gentrification indicators were similarly shifted.

III. Data Exploration

A. *Overview of Taxi Activity*

The first exploration of the taxi dataset was based on descriptive analytics and spatial heatmaps with the goal of visually identifying how different dimensions of taxi activity are distributed across TLC zones and to detect spatial patterns.

Heatmaps were generated for the main indicators of taxi activity, including pickup demand, drop-off demand, average trip value, average trip distance, tip rate and WAV (Wheelchair-accessible vehicle) requests. These visualisations provide a broad overview of taxi activity throughout NYC.

Taxi activity within New York City was found to be highly uneven. Overall, there are several TLC zones which, combined, experience the majority of both pickup and drop-off activity. As shown in Figure 3 and Figure 4, these activity hotspots are typically located in major transport hubs and commercial areas, although the pickup and drop-off activity is almost the same within each of these zones. However, zones with higher average trip frequencies and trip distances tend to be located in areas such as airports and, within these zones, taxis receive tips and WAV requests for wheelchair passengers more often.

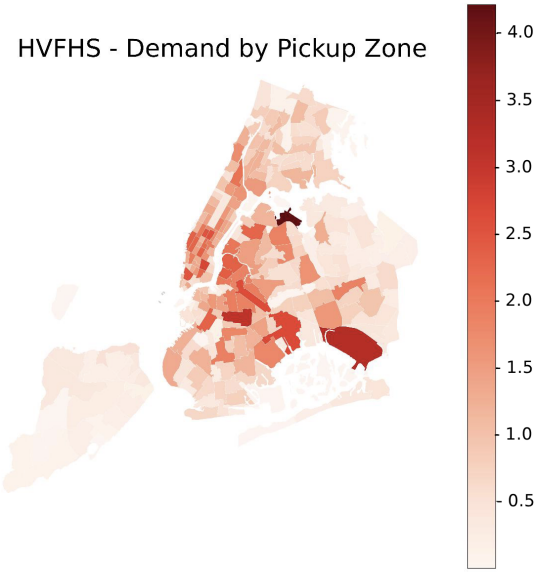


Figure 3: Spatial distribution heatmap of pickup demand throughout TLC zones.

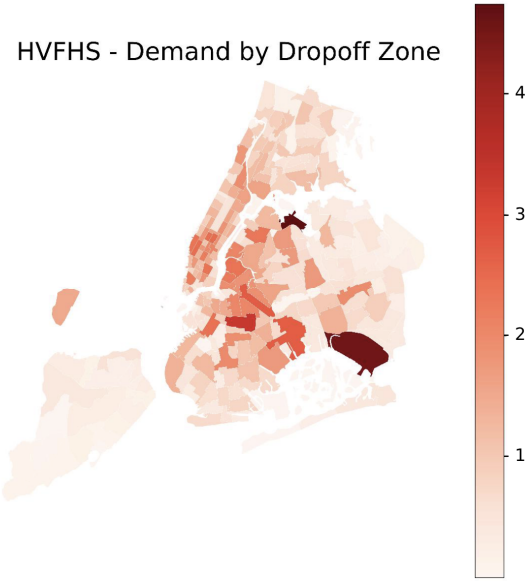


Figure 4: Spatial distribution heatmap of drop-off demand throughout TLC zones.

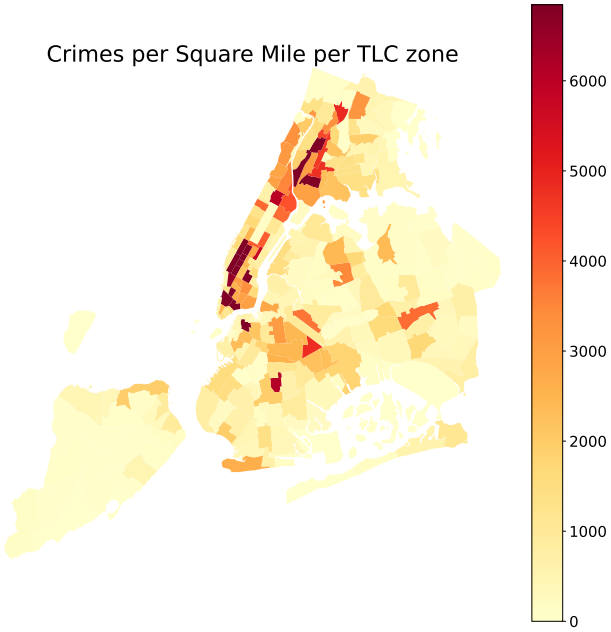


Figure 5: Heatmap showing the frequency of total crimes across TLC zones (capped at 95th percentile).

B. Initial Exploration of Crime Patterns

The initial exploration of the dataset aimed to identify how arrests were distributed across TLC zones and whether these patterns overlapped with taxi activity. Total arrest counts were first mapped across all zones, capped at the 95th percentile to avoid undue compression of colour scales in the heatmap. As shown in Figure 5, the crime density is most concentrated in centralised areas such as central and Lower Manhattan. Alongside this, hotspots are also present across the Bronx. These areas closely mirror the high taxi-demand zones identified in Section III-A. Further heatmaps were produced for offences relevant to the car-for-hire industry (such as carjacking, reckless driving and traffic violations) to test whether zones with stronger HVFHV presence also exhibited higher levels of these crimes. White-collar and blue-collar crimes were separated to better understand whether different crime types had different spatial distributions in relation to taxi activity.

C. Initial Exploration of Gentrification Indicators

Heatmaps were created to map the change in the indicators gathered across 2019 and 2025. Calculation of an overall gentrification score was subsequently attempted by combining each measure into a normalised score, with an even contribution of the collected metrics (all normalised). However, there was difficulty in assigning each metric a weight with potential for unique patterns to be "drowned out" in each metric. As it did not produce consistent results, this approach was abandoned. Consequently, it was decided to analyse and compare the TLC taxi data to each gentrification measure individually. Therefore, the remainder of this investigation will be in the modelling section.

IV. Data Modelling

A. Fare and Tip Prediction

This section describes the modelling approaches we used to analyse the cleaned HVFHV dataset. Several tasks were approached, including fare and tip prediction. The dataset was split into training (70%), validation (15%) and test (15%) sets. Hyperparameter tuning was carried out on validation data on a subset of the full dataset to reduce computational cost, with final evaluation performed on the test set. Model performance was assessed using appropriate metrics for each task. The regression model was evaluated using metrics such as mean absolute error, whilst classification models used measures such as accuracy and F1 score.

1) Fare Prediction: Fare prediction was approached in two ways: as a regression problem to predict the exact value of the fare and as a classification problem where fares were grouped into fare bands. XGBoost, a gradient boosting method, was chosen for both tasks given the likely non-linear relationships between trip features and fare outcomes.

The target variable for both approaches was `base_passenger_fare`. The feature set is summarised in Table II. These features were chosen to capture journey characteristics (distance and duration), temporal variation (hour and day), pricing adjustments (fees and surcharges) and spatial or provider-level effects. Pickup and drop-off location were removed from the fare models, since they would indirectly encode journey distance and duration instead of adding more information.

Basic feature engineering was applied by extracting temporal variables (hour of day and day of week) from the pickup timestamp.

Table II: Features used for fare prediction.

Category	Features
Trip characteristics	trip miles, trip time
Temporal features	hour of day, day of week
Pricing components	airport fee, congestion surcharge, CBD fee
Provider information	provider, shared trip flag

Hyperparameter tuning was carried out using a fixed validation split, with the search performed on a five-million row sample to reduce computational cost. The best-performing configuration was then retrained on the combined training and validation data before final evaluation on the held-out test set. The selected hyperparameters are shown in Table III.

Table III: Best hyperparameter configurations for XGBoost models.

Parameter	Fare Regression	Fare Classification	Tip Classification
Number of estimators	300	300	1500
Maximum depth	10	10	8
Learning rate	0.03	0.03	0.05
Subsample	0.8	0.8	0.8
Column sample by tree	0.8	0.8	0.8

For the regression task, performance was evaluated using mean absolute error (MAE), which provides a measure of average prediction error. For the classification task, fares were grouped into three classes as shown in Table IV, and performance was evaluated using accuracy and F1 score.

Table IV: Fare classification classes.

Class	Fare Range
Low	\$0 – \$15
Medium	\$15 – \$40
High	\$40+

2) Tip Prediction: Tip prediction was formulated as a classification task only. As a large proportion of trips record zero tips, exact value prediction is unstable and a binary formulation (whether a tip occurred at all) was adopted instead. Specifically, a trip was labelled as `tip_happened = 1` if `tips` > 0, and `tip_happened = 0` otherwise.

The target variable was derived from the original `tips` field. The set of features used for this task is summarised in Table V. These features were selected to capture trip characteristics, pricing structure, spatial variation and temporal effects; all of which are expected to influence tipping behaviour.

Table V: Features used for tip prediction.

Category	Features
Trip characteristics	trip miles, trip time, speed
Pricing variables	base fare, tolls, taxes, fees, total charge
Derived metrics	fare per mile, fare per minute
Temporal features	hour, day of week, month, weekend
Location	pickup zone, drop-off zone
Service information	provider, shared trip, accessibility flags

Feature engineering included extracting temporal variables (hour, day of week and month) from the pickup timestamp, creating a weekend indicator and constructing derived features such as fare per mile, fare per minute, trip speed and total charge (excluding tip).

Class imbalance was handled using the `scale_pos_weight` parameter in XGBoost, calculated from the ratio of non-tipped to tipped trips in the training data. This was necessary as non-tipped trips form the majority class.

An XGBoost classifier with a binary logistic objective was used for this task. This model was chosen due to its strong performance on large tabular datasets and its ability to capture non-linear relationships between variables such as cost, trip characteristics and location.

Hyperparameter tuning was performed using random search over a predefined parameter space, with each configuration evaluated on the validation set. This approach was used instead of grid search due to the size of the dataset, allowing a wider range of configurations to be explored efficiently. The best-performing configuration was then selected based on validation performance and evaluated on the test set. The default configuration used when tuning was not applied is shown in Table III.

During training, early stopping was applied using the validation set. Predicted probabilities were generated and the final decision threshold was selected based on validation performance to maximise F1 score, rather than using the default threshold of 0.5.

Model performance was evaluated using multiple metrics, including ROC AUC, F1 score, precision, recall and accuracy. The F1 score was particularly important due to class imbalance, as it balances precision and recall.

B. Crime Analysis

The model was constructed to determine whether there is a relationship between taxi activity in New York City and arrests that occurred within the city's TLC zones. Taxi activity is represented by the number of trips by taxi, collected from the HVFHV dataset, while crime activity is represented by arrests that occurred within each zone, collected from the NYPD database and previously matched to each zone by its `LocationID`.

Separately for each TLC zone, taxi activity was measured over a monthly-time series and by TLC Zone using either pickup or drop-off trips from the HVFHV dataset. Additionally, arrests within each zone over time were collected by zone using the `OFFENSE` category descriptions (`OFNS_DESC`). These two tables were merged on time and `LocationID`, allowing the two datasets to be compared within each zone.

Within the combined tables of each TLC zone over time, Pearson's correlation coefficient was calculated between taxi activity and arrest counts. This was computed both globally across all zones for each offence category and locally within individual TLC zones, to identify whether any zone-specific relationships existed. Each calculation also included the associated p-value and number of observations used, with correlations derived from too few observations discarded prior to interpretation.

Finally, within the merged tables of each TLC zone over time, heatmaps (Figure 6 & Figure 7) were created that displayed the number of trips made by taxis and the number of crimes that occurred in each TLC zone. These maps depicted the relationship between taxi activity and crime activity within each TLC zone.

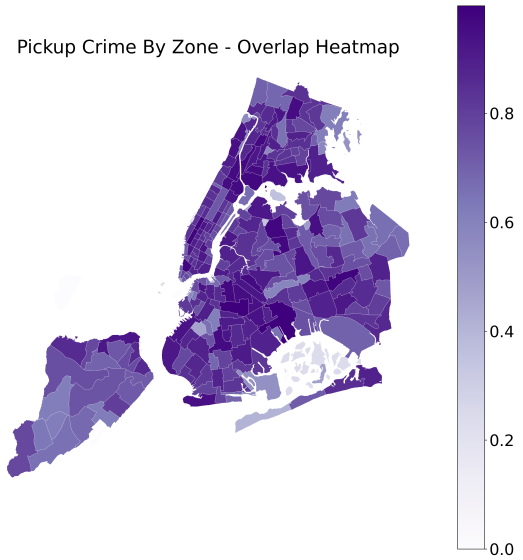


Figure 6: Overlap heatmap between crime frequency and pickup demand.

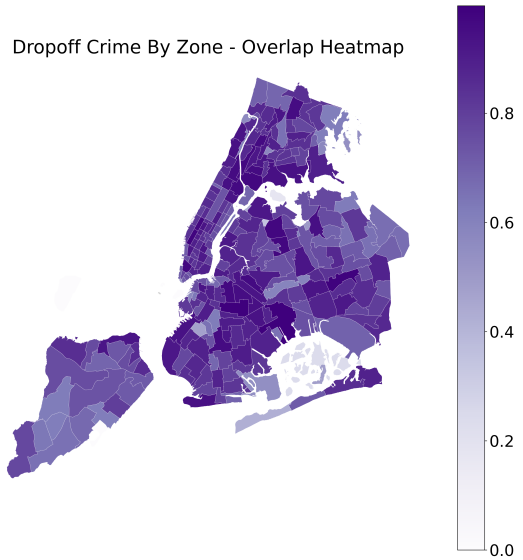


Figure 7: Overlap heatmap between crime frequency and drop-off demand.

C. Gentrification Analysis

In order to compare with gentrification measures, the TLC data had to be collated into a series of indicators to facilitate easier comparison and analysis. A score was created for each zone combination, composed of two primary metrics: trip volume growth (the percentage change in total pickups and drop-offs from a given zone) and revenue growth (the percentage change in total driver pay) between a base year and a target year. This was also normalised to make sure it could be compared. As a result, a value was calculated between all available zone pairings, as follows: $N = \binom{264}{2} = 34,716$.

The most significant scores from this set were then examined and analysed for a relationship with the socioeconomic indicators of gentrification taking place across the same zones. However, this approach did not show any significant correlation, perhaps because the raw data was hidden behind too much abstraction.

Upon further testing, the best results were obtained by directly analysing an individual gentrification metric with a raw TLC metric for a given zone.

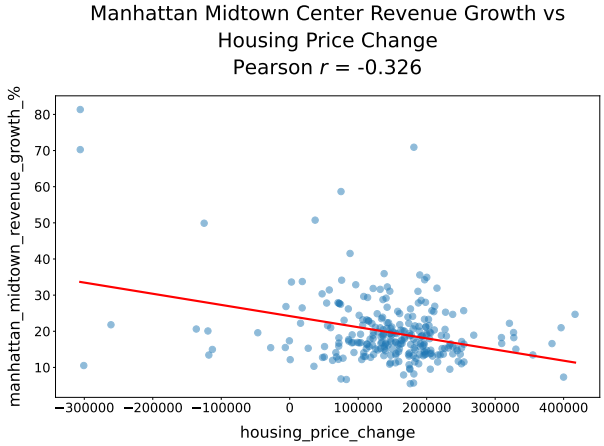


Figure 8: Scatterplot between the housing price change and taxi revenue growth.

An example of this is seen in Figure 8. Though not an overall strong pattern, it does suggest there is a slight relationship between a reducing taxi revenue growth and significant housing price change.

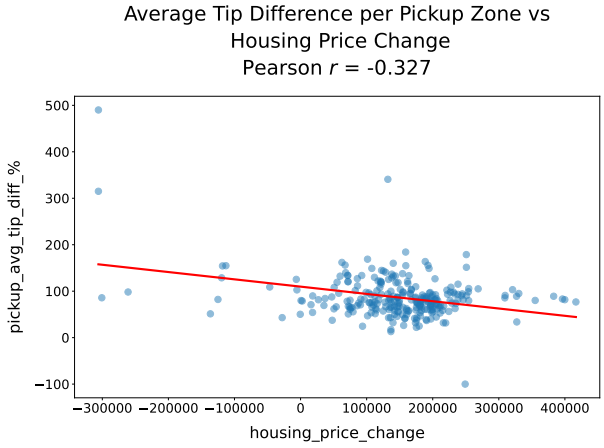


Figure 9: Scatterplot between the housing price change and the tip increase per pickup zone in the same area.

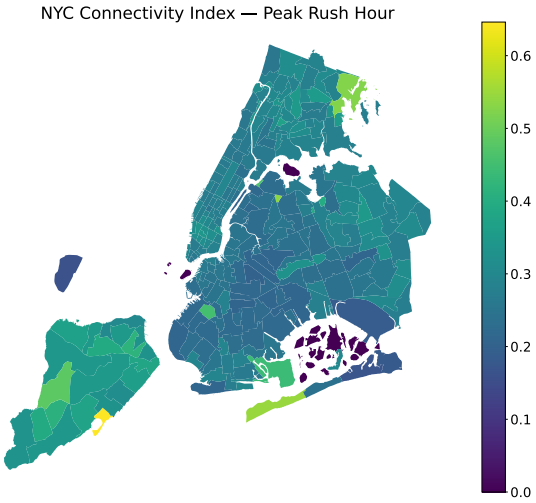


Figure 10: Heatmap portraying the connectivity of each TLC zone in NYC based on travel times during morning rush hour.

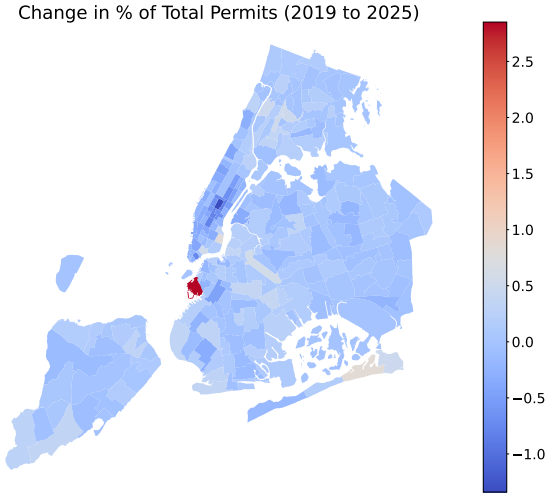


Figure 11: Heatmap displaying the zonal percentage change in proportion of issued building permits between 2019 and 2025.

Across other especially central zones, a similar weak effect was seen as shown in Figure 9.

The Pearson correlation was chosen for this analysis as it provides a standardised measure of the strength and direction of the linear relationship between two continuous variables, which is ideal for this scenario.

By producing a value between -1 and 1, it allows for a direct comparison between Midtown and other zones to see if the "gentrification signal" is uniform across different urban environments. Beyond such a relationship in the centralised areas, correlations were non-existent in other areas. The implications of this are further discussed in the analysis section.

Building permits were repositioned to fall within the TLC zones as they are stored as coordinate and address data. Between years 2019 and 2025, permit applications dropped significantly across the whole of NYC. Thus, the number of permits issued per TLC zone was treated as a proportion of the total number of issued permits within a certain year. It was then the change in these proportions which was evaluated, as seen in Figure 11.

To further characterise zone-level economic activity, a connectivity index was derived from inter-zone taxi travel time, using averages of inter-zone journeys of morning commuters (between 07:30 and 08:00) across 2025. Outliers were removed and zones with fewer than 50 trips between them were not counted. These contributed to a weighted inverse travel time metric for each zone, favouring frequently travelled, fast routes. Connectivity using this metric across NYC can be seen in Figure 10.

V. Findings

A. Fare and Tip Prediction Results

This section evaluates the performance of the XGBoost models for fare and tip prediction against the baselines in Table VI. The baseline classification models used a random classification, whereas the fare prediction baseline used the median value of the dataset. Table VII shows the performance of our proposed models which indicates that our models perform better than random selection.

Table VI: Baseline model performance.

(a) Fare Regression (Median)

Metric	Value
MAE	14.82
RMSE	26.85
R^2	-0.08

(b) Classification (Random)

Task	Accuracy	F1-score
Fare	0.38	0.33
Tip	0.69	0.19

Table VII: Model performance improvement over baseline.

(a) Fare Regression

Metric	Value
MAE	5.09 (-9.73)
RMSE	9.02 (-17.83)
R^2	0.82 (+0.90)

(b) Classification

Model	Accuracy	F1-score
Fare	0.84 (+0.46)	0.84 (+0.51)
Tip	0.61 (-0.08)	0.29 (+0.10)

The fare regression model achieves a strong performance, with an MAE of 5.09, an RMSE of 9.02 and an R^2 score of 0.82 in Table VII. Compared with the baseline in Table VI, this represents a substantial improvement in predictive performance. The gap between MAE and RMSE suggests that the

model performed well on most trips, while a smaller number of atypical higher-fare journeys contributed disproportionately to the largest errors.

The fare classification model also achieves a strong performance, as shown in Table VII, with an overall accuracy and F1 score of 0.84. Table VIII demonstrates that the model performed best on the low and medium classes, while the high-fare class proved slightly harder to detect, with recall dropping to 0.75. This pattern is also visible in Figure 12, where a significant number of misclassifications can be seen for the true high class.

Table VIII: Class-wise precision, recall, and macro F1-score for fare-band classification.

Fare Band	Precision	Recall	F1-score
Low	0.8830	0.8651	**0.8740**
Medium	0.8131	**0.8731**	0.8420
High	**0.8845**	0.7477	0.8103

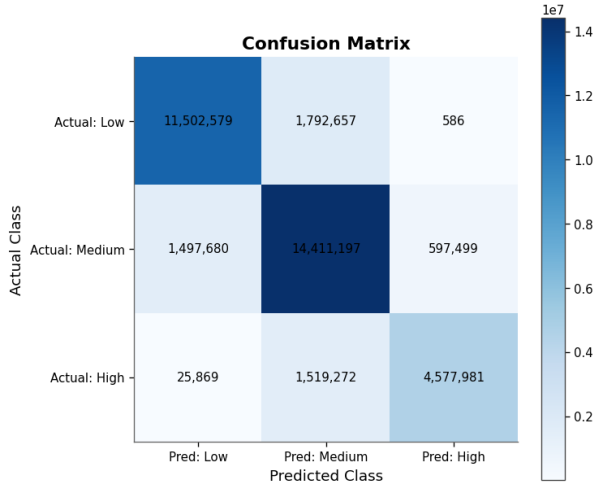


Figure 12: Confusion matrix for the XGBoost fare-band classifier.

Tip prediction proved to be much harder than the fare prediction. The tip classifier achieved an F1 score of 0.29 (Table VII) which is an improvement from the baseline F1 score of 0.19 in Table VI. This means that the model did learn some useful signals for identifying trips with tips, but the overall performance remained limited. The achieved accuracy of 0.61 (Table VII) is lower than the baseline accuracy (Table VI) and is not an informative metric here as there is significant class imbalance, with only approximately 18.75% of trips including a tip, meaning a naive zero-tip model would achieve an accuracy of around 81.25%.

Even though the model captures some signal in tipping behaviour, predicting tips remains a challenging task due to class imbalance and the variability in passenger behaviour.

A key source of this noise is the under-reporting of tips in the HVFHV dataset: only in-app trips are captured and trip records are submitted electronically via internal platform APIs. Because cash tips are generally preferred by drivers due to their immediate availability, with no processing fees, they are entirely absent from the dataset. Therefore, a significant portion of zero-tip labels are likely mislabelled, and the model may be trained on a corrupted target variable. This label noise likely reduces the achievable performance of the model, as it obscures the true tipping behaviour and makes the classification task inherently more difficult. Figure 13 also shows that the model struggles with predicting true positives.

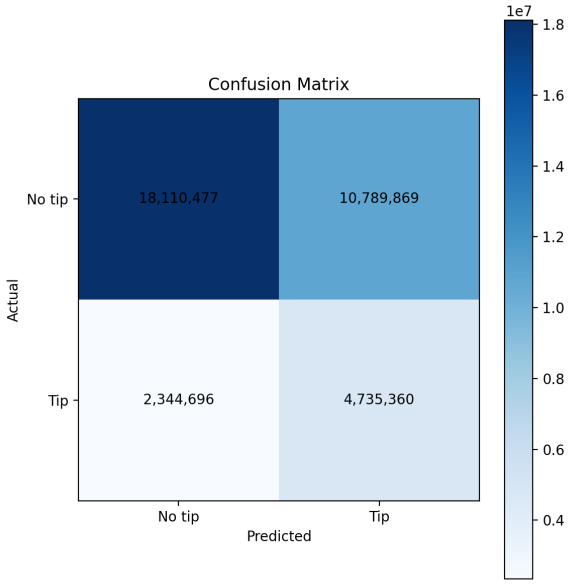


Figure 13: Confusion matrix for the XGBoost tip classifier.

Figure 14 shows that fare prediction was driven mainly by trip distance and duration. In the fare regression and classification model, the most important features were `trip_miles`, `trip_time` and `cbd_congestion_fee`.

Tip prediction, on the other hand, relied on a different set of features, the most important of which were `airport_fee`, `congestion_surcharge`, `shared_request_flag` and `cbd_congestion_fee`. This suggests that tipping behaviour is influenced by the trip context on top of the length of the journey.

B. Crime Analysis Results

The results of the analyses indicate a positive relationship between taxi activity and crime across the NYC TLC zones. The majority of the crime categories exhibited a moderate positive correlation to both pickup and drop-off demand for taxis, indicating that zones with high demand for taxi rides experience high counts of arrests within those zones (Figure 15). The strongest correlations between crime and taxi movement were found for categories of crimes like Disorderly Conduct, Criminal Mischief and Assault-Related offences. Furthermore, drop-off demand for taxis had stronger correlations to crime categories than taxi pickup demand.

The comparison between broad crime groups further supports the positive relationship interpretations. As shown in Table IX, taxi activity was more correlated with non-white-collar

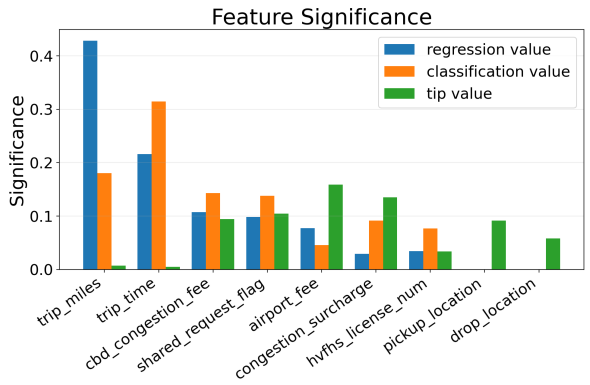


Figure 14: Top feature importances for fare and tip XGBoost models.

crimes ($r = 0.417$) than with white-collar crime ($r = 0.333$), suggesting that taxi demand is more closely associated with street-level and public-space offences than with crimes less dependent on physical urban concentration.

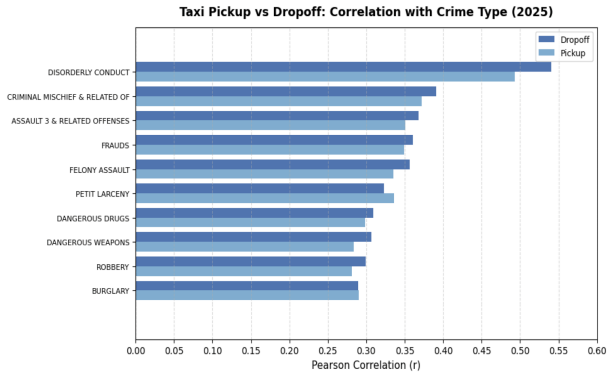


Figure 15: Correlation between taxi activity and crime across TLC zones.

Table IX: Correlation between taxi activity and broad crime categories.

Crime category	Pearson correlation (r)
Taxi vs White Collar Crime	0.333
Taxi vs Non White Collar Crime	0.417

C. Gentrification Results

Figure 11 clearly shows that, for building permits between 2019 and 2025, focus has shifted markedly away from Midtown Manhattan and towards Red Hook, Brooklyn. This aligns with widely reported gentrification trends in Red Hook throughout this time period [13][14]. The zones covering Rockaway Beach also experienced a slight increase in permit share, following similarly observed trends in recent years [15]. These shifts in property investment broadly correspond to increases in total trip volumes in these TLC zones, suggesting that a relationship exists between them in areas undergoing gentrification.

Figure 8 and Figure 9 may point to a saturated market where rising property values are often driven by luxury residential investments. These units might not necessarily even be inhabited all the time. As a result, this could have a slight negative effect on taxi revenues as wealthy residents do not live there. Another explanation could be increasing tourist rates in the area. Tourists are more likely to use the examined Uber or Lyft vehicles, unlike wealthy residents who might have personal drivers or different means of transport. As the housing prices also increase, a demographic change might also be occurring. This might bring about a shift in culture and, thus, less tipping. More research into demographic trends would need to be done to confirm this.

The connectivity index (Figure 10) shows that Staten Island and lower parts of Manhattan are among the highest scorers for the least travel time to other zones during peak morning rush hour, reflecting fast and high-frequency travel to (and, to a lesser extent, from) NYC's main areas of employment. This is consistent with the idea that taxi-accessible areas attract the most infrastructure investment.

The lack of a correlation for the gentrification data in less centralised areas suggests a higher reliance on public transport, walking or personal vehicles despite potential gentrification occurring. In lower-income or middle-class zones, changes in education or income might show up as increased subway usage or car ownership rather than taxi trips. This makes taxi data a "biased" proxy towards richer areas.

In conclusion, this demonstrates that whilst serving as a useful proxy for some human activity, taxi activity does not account for all aspects of it due to other confounding variables.

VI. Discussion and Conclusion

If this study were to be done again, a more thorough longitudinal study for the gentrification axis could be done. As we have found out, it is not a process easily captured via the proxies in the timescales we have used. Other more complex gentrification measures could also be included such as mortgage rates, rent rates per each area, volume and approval rates of mortgages, a shift from "blue-collar" or service jobs to "white-collar" professional and managerial positions or ethnic composition change. However, the scope of this study only allowed for a limited selection of indicators. It should be noted that, whilst some correlations have been observed, this does not imply any causal definition to our findings. Moreover, Metropolitan Transport Authority (MTA) data could be analysed in parallel with TLC data. This could give a more detailed view of the kinds of factors that drive gentrification.

A further limitation of this approach to analysing human patterns across NYC is that HVFHVs are not typically used by lower-income residents. As outlined in other studies [16], such residents often become invisible to big data. In many cases, NYC "dollar vans" are used as an alternative form of transport. As these are primarily cash-based and do not generate GPS data, it is more difficult to implement this approach.

In addition, our findings related to the correlation between disorderly conduct and taxi presence have been further validated by and explored in related literature and policy research. The NYC Office of Nightlife mentions such strategies for Integrated Nightlife Management [17]. It also highlights the importance of coordination with TLC for a steady supply of for-hire vehicles outside nightclubs. Specifically, the lack of reliable transportation between 2AM – 4AM around such hotspots causes loitering friction, underlining the potential benefit that analysing links between taxi activity and criminal behaviour brings.

Furthermore, the importance of the airport fee to the overall price of a given ride has been highlighted as a useful policy lever, serving to keep airport taxi fares high, both to help alleviate congestion and to encourage the use of public transport (such as the AirTrain at JFK) [18]. This report specifically models an increase in the airport fee to test any effects on the AirTrain ridership, showing the importance of leveraging the main components of a taxi ride price to create more sustainable cities. However, it has been suggested that increasing the taxi airport fee leads to higher overall prices for casual/low-income travellers who might need the door-to-door service for safety or luggage requirements [19]. As a result, a two-tier access system is created where the wealthy pay for convenience and the poor are forced into longer, multi-modal commutes. Consequently, a successful policy must analyse the effects of introducing such fees for equal access for everyone.

As described in section V, question (1) is answered by showing a moderate positive correlation between taxi drop-off demand and disorderly conduct arrests across TLC zones. For question (2), fare prediction achieved strong performance, whereas tips are generally less reliable since the dataset is zero-biased and does not include cash tips. Finally, question (3) was partially answered by showing some relationships between housing prices and taxi presence in specific zones, which is consistent with independent observations of gentrification in those areas.

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VII. Appendix

Supplementary figures and visualisations that were generated during the course of this project but not included in the main report are available here: Appendix Materials [Online]